

Philosophy 130: Philosophy of Space and Time
Assignment 5: Simultaneity and the Clock Hypothesis

Due: May 12, in class

Name: _____

Please draw and write your answers in the space provided below, and on an additional sheet of paper (for Question 1, which requires a clearly drawn spacetime diagram).

Question 1. A pole approaches a barn at high relativistic speed. The pole's rest length is greater than the barn's rest length. The barn has open doors at its near end and at its far end. Two observers describe the encounter: Beth, at rest in the barn's frame, and Paul, at rest in the pole's frame.

The pole is moving fast enough that, in Beth's frame, it is Lorentz-contracted to be just shorter the barn's length. So according to Beth there is a moment when the front of the pole is at the far door and the back of the pole is at the near door — the pole almost fills the barn. According to Paul, however, the moving barn is Lorentz-contracted; the (rest-length) pole is longer than the contracted barn, and the pole cannot fit inside the barn at any moment.

Two events define the puzzle:

- **Event A:** the front of the pole is at the far door.
- **Event B:** the back of the pole is at the near door.

(a) In Beth's frame, are events A and B simultaneous? Briefly say why — and explain how this corresponds to her saying "the pole fits inside the barn."

(b) In Paul's frame, are events A and B simultaneous? State which event Paul takes to occur first, and explain how this corresponds to his saying "the pole does not fit inside the barn at any moment."

(c) Sketch a spacetime diagram in Beth's (the barn's) frame. Include:

- the vertical worldlines of the two barn doors;
- the two parallel tilted worldlines of the front and back of the pole — together forming a tilted strip (the pole's *worldsheet*);
- events A and B, marked clearly;
- a horizontal line representing Beth's simultaneity slice that passes through both A and B;
- a tilted line representing Paul's simultaneity slice through Event A.

Your diagram should make visible that Paul's simultaneity slice through A does *not* pass through B.

(d) Use your diagram to explain in 3–4 sentences why each observer is correct in their own frame. Both observers describe the same set of events; what they disagree on is which slicing of spacetime counts as "at one moment." Each is taking a different cross-section of the same pole worldsheet.

Question 2. What do clocks measure?

(a) In **Galilean spacetime**, time has an absolute structure. Spacetime is foliated into *instants* — hyperplanes that extend across all of space at a single time. Briefly explain what clocks measure in Galilean spacetime, and compare the results for two clocks that move along different worldlines, and comment on why this is the only plausible answer given the structure.

(b) Now consider the contrasting situation in Minkowski spacetime. In **Minkowski spacetime**, by contrast, there are no absolute instants — no single foliation of spacetime that all observers must agree on. Why is the question of what a clock measures a substantive empirical question in this context? What would we need to rule out to show that the clock hypothesis holds? (Consider, for example that a clock's rate could in principle depend on various features of its worldline — for example, on its acceleration, on its recent history, etc.)

(c) Identify one empirical phenomenon, from lecture or reading, that supports the clock hypothesis. Briefly say how the phenomenon counts as evidence.